

APM 1205 – Applied Statistical Modeling

Formative Assessment 1: Examining Data, Transformation, and Regression

Introduction

This report examines the 2007 country-level Gapminder data and studies the relationship between life expectancy at birth (lifeExp) and GDP per capita (gdpPercap). The response variable is life expectancy, while GDP per capita is the primary explanatory variable. The analysis first examines the distributions and bivariate relationship, then applies a natural-log transformation to GDP per capita and compares two simple linear regression models.

The documentation describes the teaching dataset as containing 142 countries observed every five years from 1952 to 2007, with variables including country, continent, year, life expectancy, population, and inflation-adjusted GDP per capita, where in this assessment uses the 2007 observations.

Data Examination

Number of observations: 142 countries.

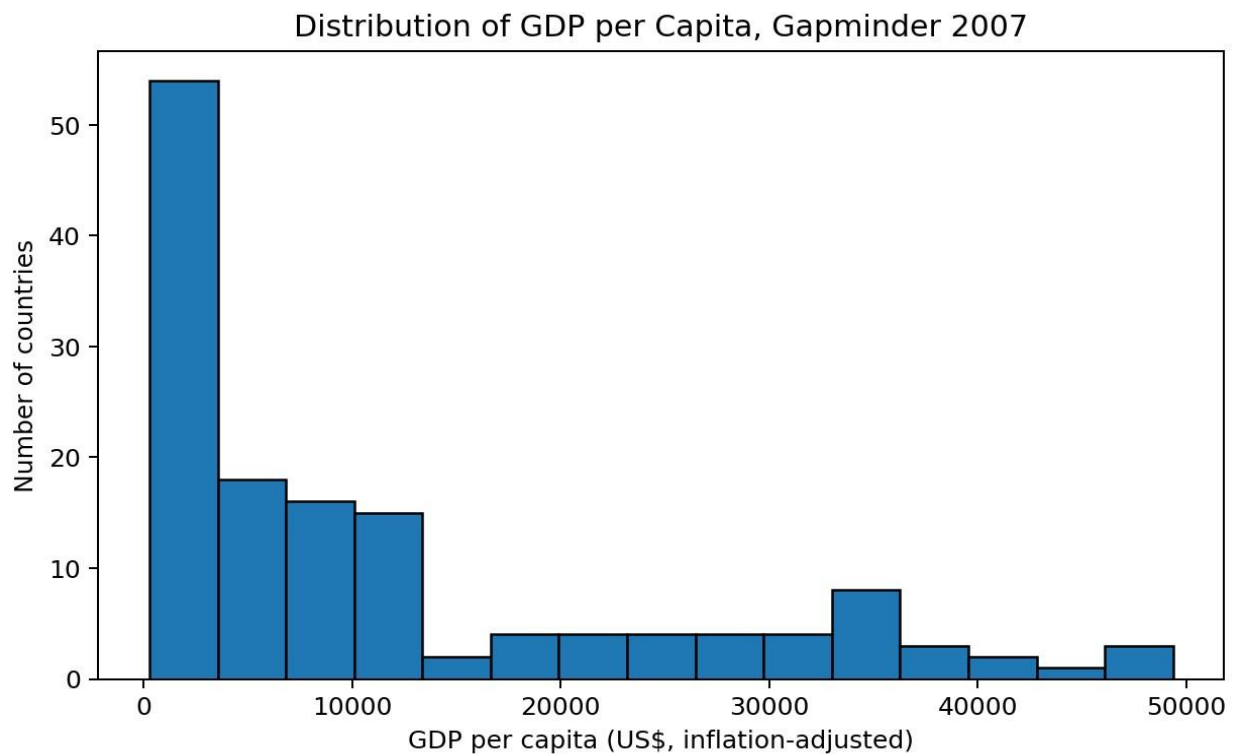
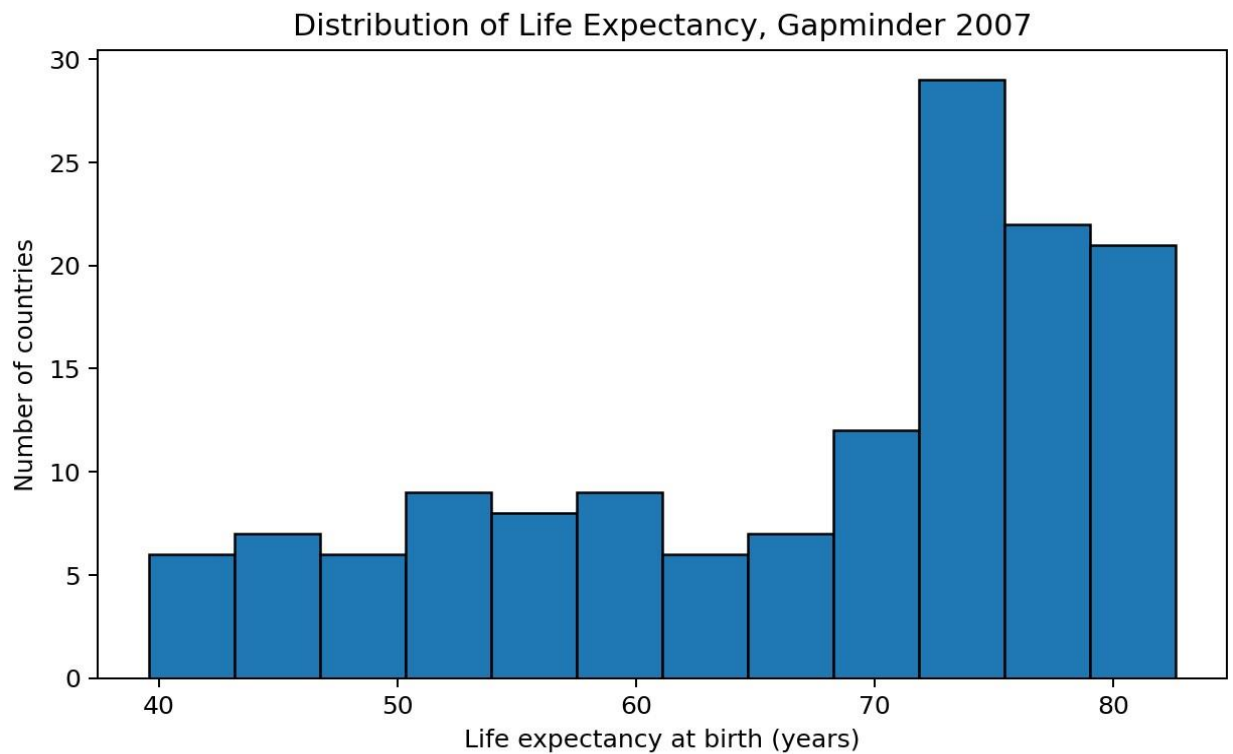
Number of original variables: 6, which are country, continent, year, pop, lifeExp, and gdpPercap.

Missing values: none in the 2007 subset.

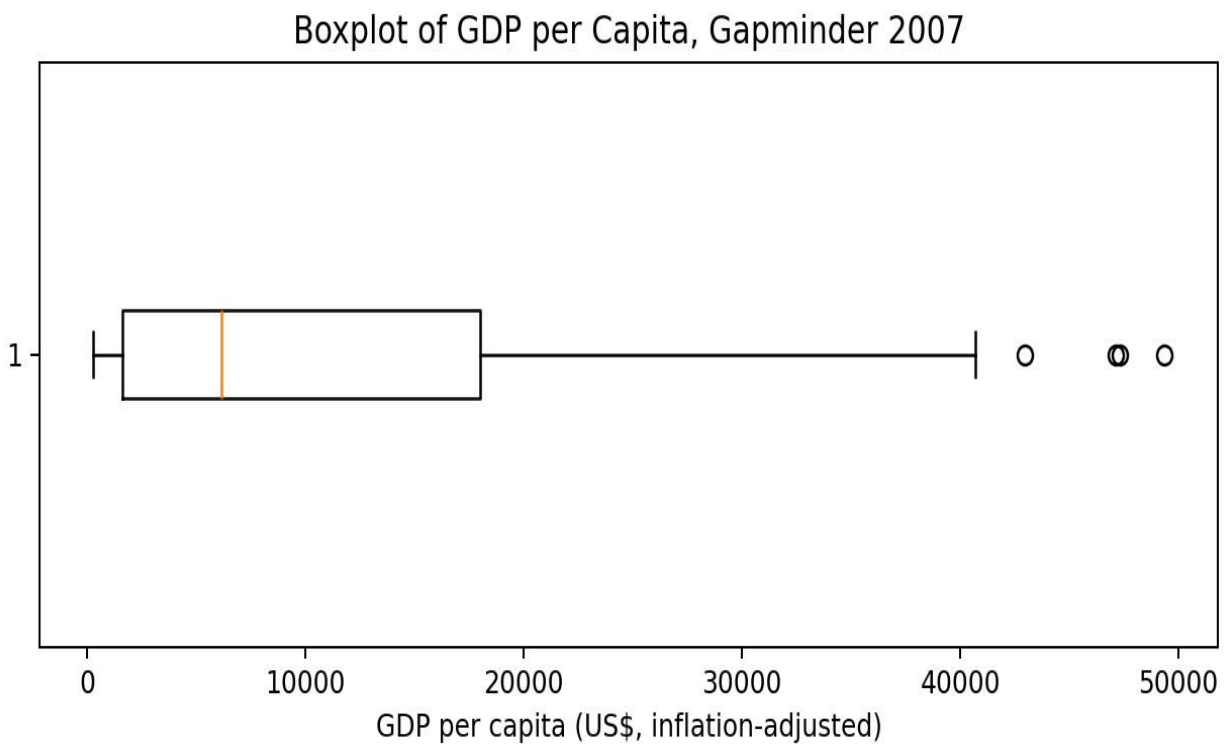
Measurement roles: lifeExp is quantitative and serves as the response variable; gdpPercap is quantitative and positive and serves as the explanatory variable.

Statistic	lifeExp	gdpPercap
Mean	67.007	11,680.072
SD	12.073	12,859.937
Min	39.613	277.552
Q1	57.160	1,624.842
Median	71.935	6,124.371
Q3	76.413	18,008.836
Max	82.603	49,357.190
IQR	19.253	16,383.993
Skewness	-0.689	1.224

Numerical interpretation. Life expectancy has a mean of 67.01 years and a median of 71.94 years, with an SD of 12.07 years. Its negative skewness (-0.689) indicates a longer lower tail caused by countries with comparatively low life expectancy. GDP per capita has a mean of \$11,680.07, much larger than its median of \$6,124.37, and strong positive skewness (1.224), showing a highly right-skewed distribution.

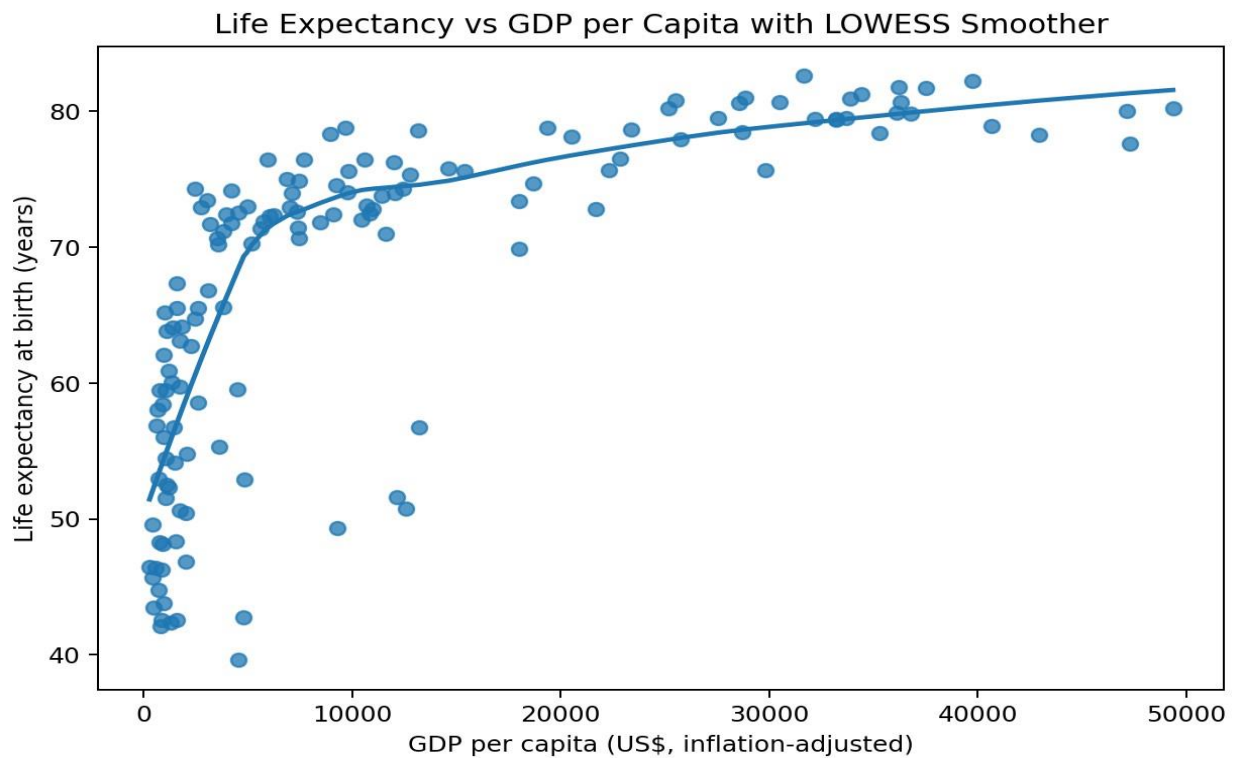


Distributional shapes. The life-expectancy distribution is moderately left-skewed and concentrated at higher values. The GDP-per-capita distribution is strongly right-skewed: many countries are concentrated at relatively low to middle values, while a small number have very high GDP per capita. The $1.5 \times \text{IQR}$ rule identifies no life-expectancy outliers. For GDP per capita, the upper fence is approximately \$42,584.83, and four countries exceed it: United States, Singapore, Kuwait, and Norway.



Bivariate Relationship

The correlation between life expectancy and GDP per capita on the original scale is $r = 0.679$, indicating a positive association. The scatterplot and LOWESS smoother show that the relationship is not perfectly linear: life expectancy generally rises rapidly as GDP per capita increases from low levels, but the increase gradually flattens at higher GDP levels. The vertical spread is also not constant, and several unusual observations fall well below the main trend.



Anscombe's Quartet demonstrates that datasets can have similar numerical summaries and fitted regression lines while having very different graphical structures. Therefore, a **scatterplot** is essential for detecting curvature, unequal spread, clusters, and unusual observations that numerical summaries alone may conceal. Here, the original-scale plot reveals nonlinearity and motivates transforming GDP per capita before relying on a straight-line model.

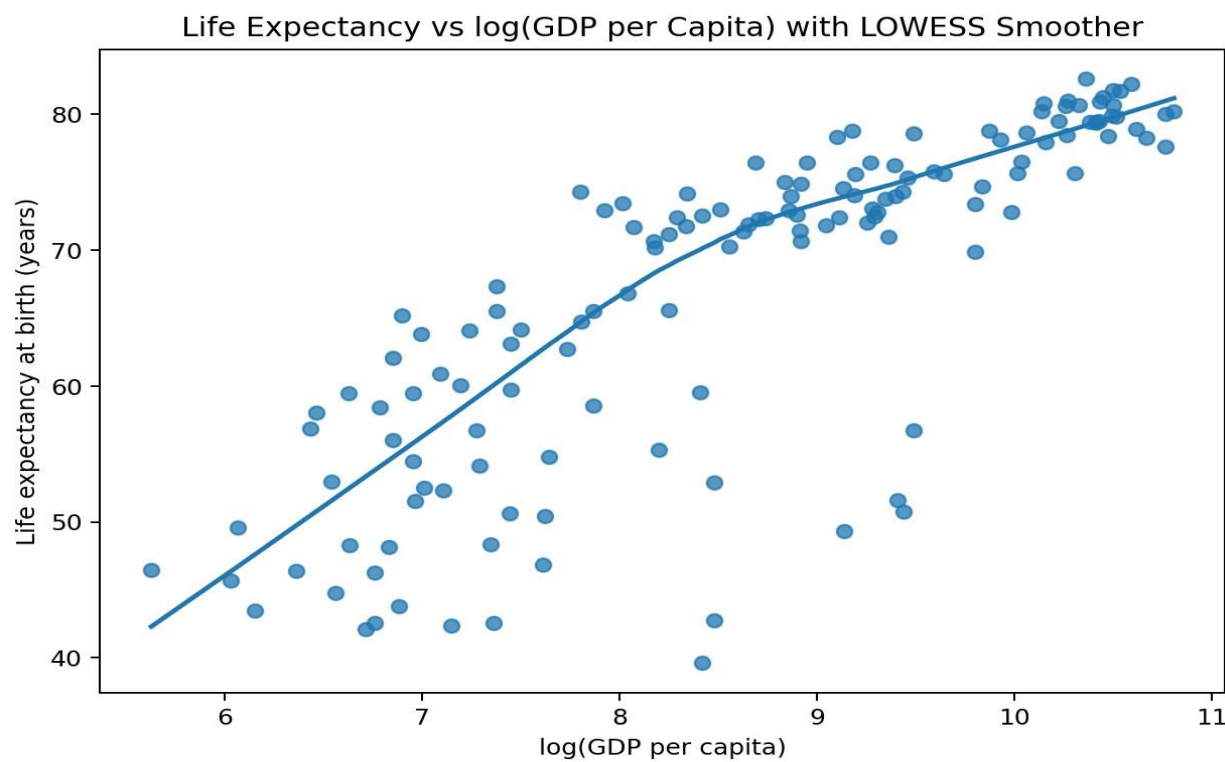
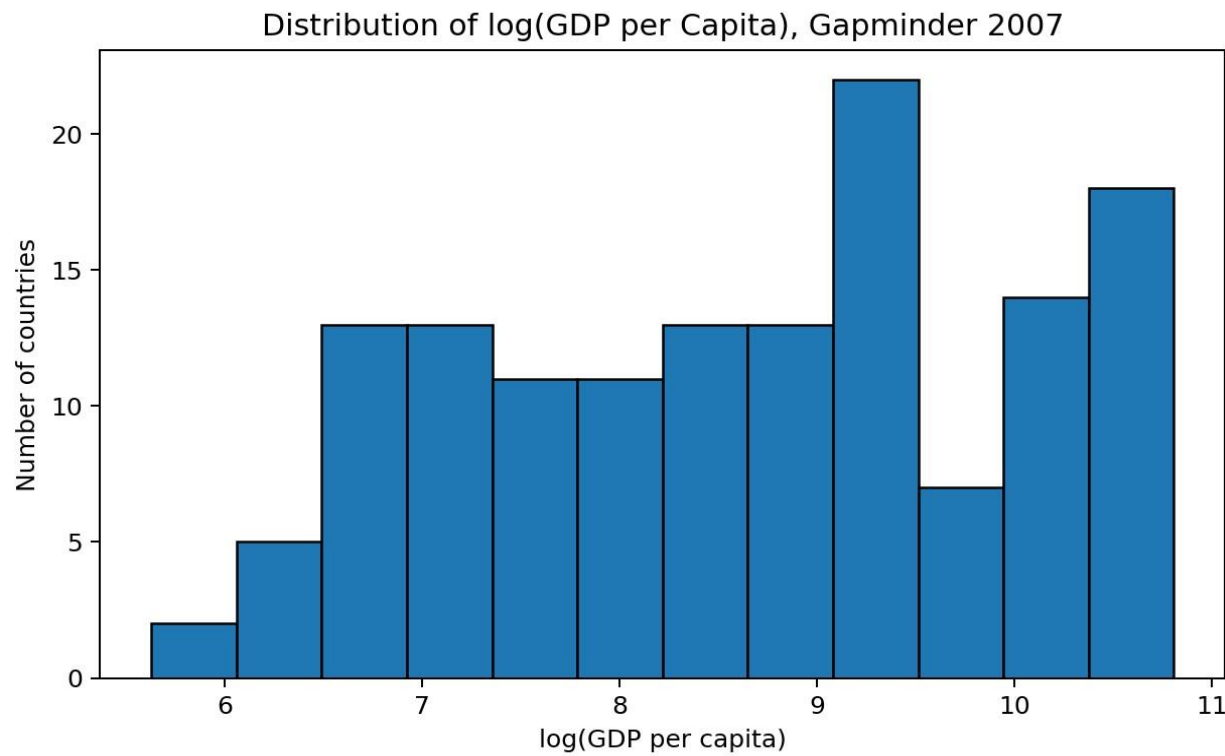
Transformation of the Explanatory Variable

GDP per capita is strictly positive and strongly right-skewed. The **natural-log transformation** compresses very large values more than small values, reduces right-skewness, reduces the leverage of extremely high GDP observations, and can make a curved relationship closer to linear. It is therefore a reasonable transformation for regression.

Statistic	Original GDP	log(GDP)
Mean	11680.072	8.616
Median	6124.371	8.720
SD	12859.937	1.356
Skewness	1.224	-0.154
Symmetry ratio	2.641	0.813

The original GDP-per-capita distribution has skewness 1.224 and a symmetry ratio of 2.641, far from 1. After logging, skewness becomes -0.154 and the symmetry ratio becomes 0.813, which is much closer to 1. Graphically,

the long right tail is substantially compressed and the transformed distribution is much more symmetric. Thus, the log transformation improves the explanatory variable for regression purposes.



Regression Comparison

Model 1: $\text{lifeExp} = 59.5657 + 0.00063713(\text{gdpPercap})$.

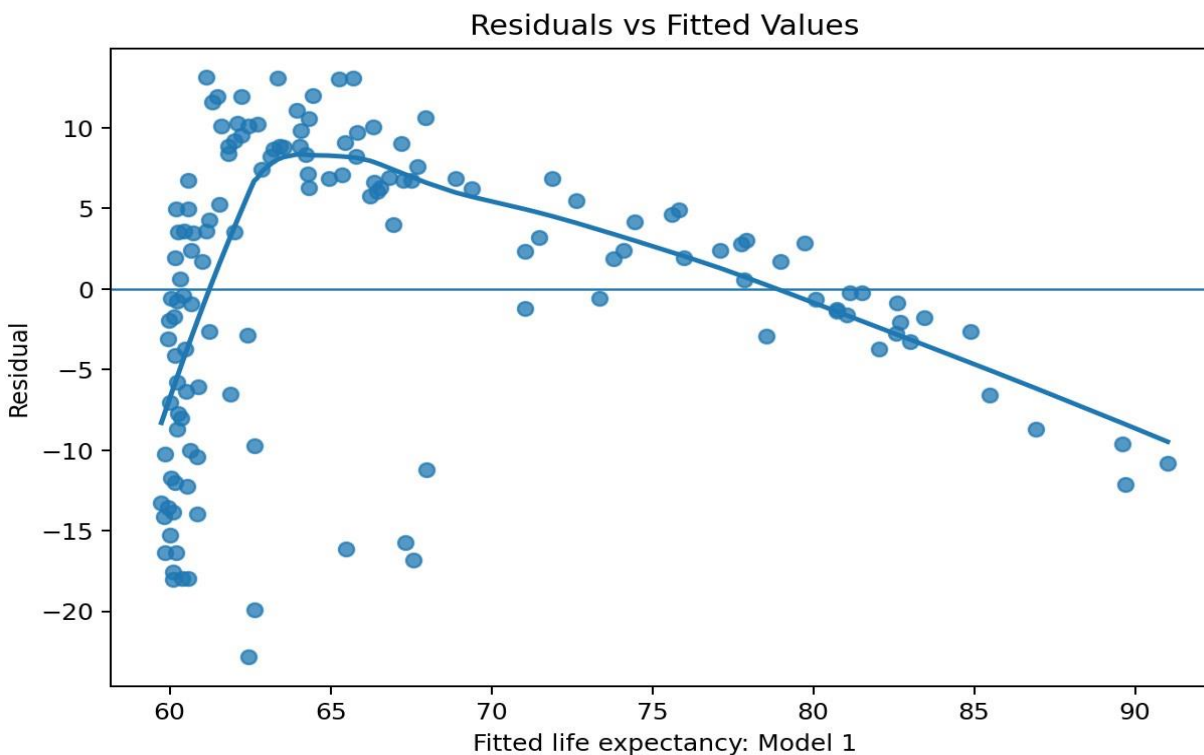
The slope means that, under this linear model, an additional \$1,000 in GDP per capita is associated with approximately 0.637 additional years of predicted life expectancy. Because the original relationship is curved, this constant-rate interpretation is not equally realistic across the full GDP range.

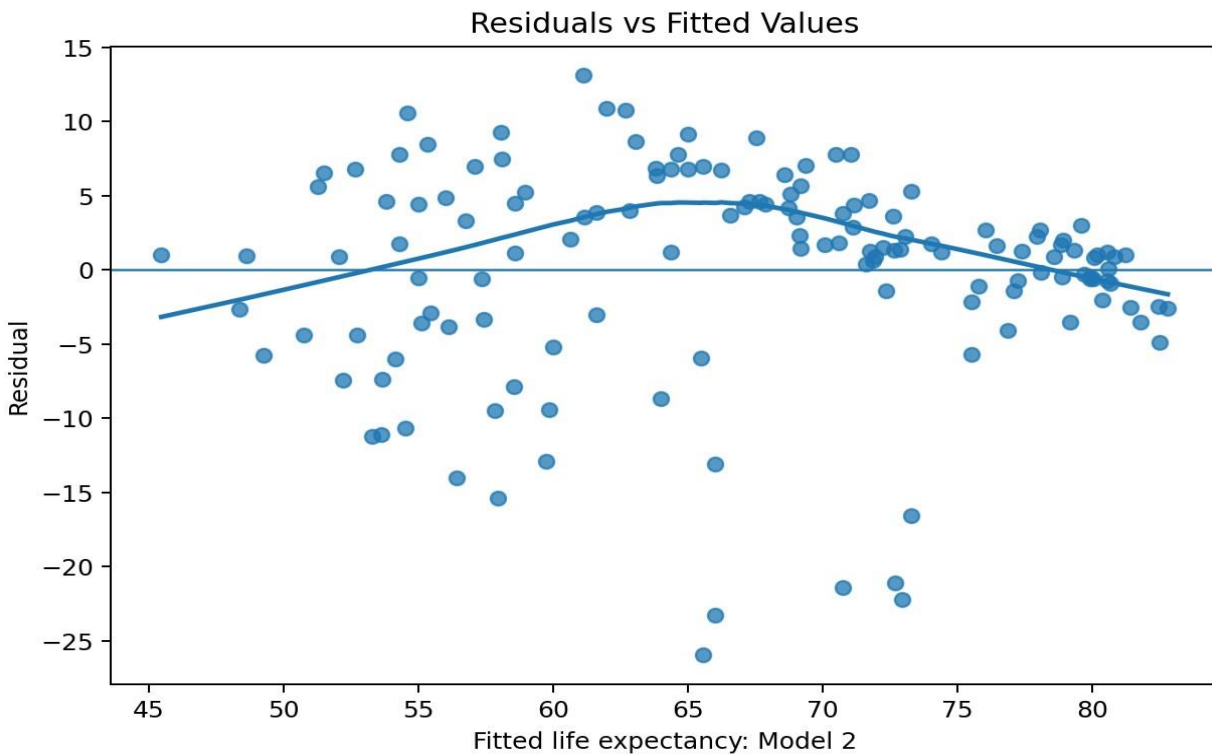
Model 2: $\text{lifeExp} = 4.9496 + 7.2028 \log(\text{gdpPercap})$.

The positive coefficient indicates that countries with higher logged GDP per capita tend to have higher life expectancy. A multiplicative increase in GDP has a constant additive interpretation on the log scale.

Criterion	Model 1: GDP	Model 2: log(GDP)
R-squared	0.4606	0.6544
Adjusted R-squared	0.4567	0.6520
Residual standard error	8.8986	7.1223

Numerical comparison. Model 2 is stronger by both requested numerical criteria: its R^2 is 0.6544, compared with 0.4606 for Model 1, and its residual standard error is 7.122, compared with 8.899. Thus, logging GDP per capita explains substantially more variation in life expectancy and produces smaller typical residual errors.





Residual evidence and model choice. Model 1 shows systematic curvature and noticeably uneven residual spread, especially because the original linear model struggles to represent the rapid increase and later flattening of the relationship. Model 2 produces residuals that are more randomly scattered around zero and has a more stable spread overall, although several countries remain unusual. Therefore, Model 2 is the more appropriate simple linear description.

Important limitation for causal interpretation. This is an observational cross-sectional comparison, so the regression association does not establish that higher GDP per capita directly causes longer life expectancy. Other factors—such as healthcare access, education, sanitation, inequality, governance, conflict, and historical conditions—may affect both economic conditions and health outcomes. Omitted variables and possible confounding prevent a causal conclusion.

Conclusion

Overall, the analysis revealed a positive relationship between national GDP per capita and life expectancy. However, the relationship on the original scale exhibited curvature, while GDP per capita showed substantial right skewness. Applying a logarithmic transformation to GDP per capita improved the symmetry of its distribution and produced a more nearly linear relationship with life expectancy. Furthermore, the transformed regression model demonstrated a higher coefficient of determination (R^2), a lower residual standard error, and more appropriate residual behavior, indicating that the log-transformed model provides a better simple regression description of the data. Nevertheless, because the analysis is based on observational data, the identified relationship should be interpreted as associational rather than causal.

References:

Gapminder - <https://raw.githubusercontent.com/resbaz/r-novice-gapminder-files/master/data/gapminder-FiveYearData.csv>

Access the github here: https://github.com/richards07-afk/LINEAR-MODELS/blob/main/LinearModels_FA1.ipynb